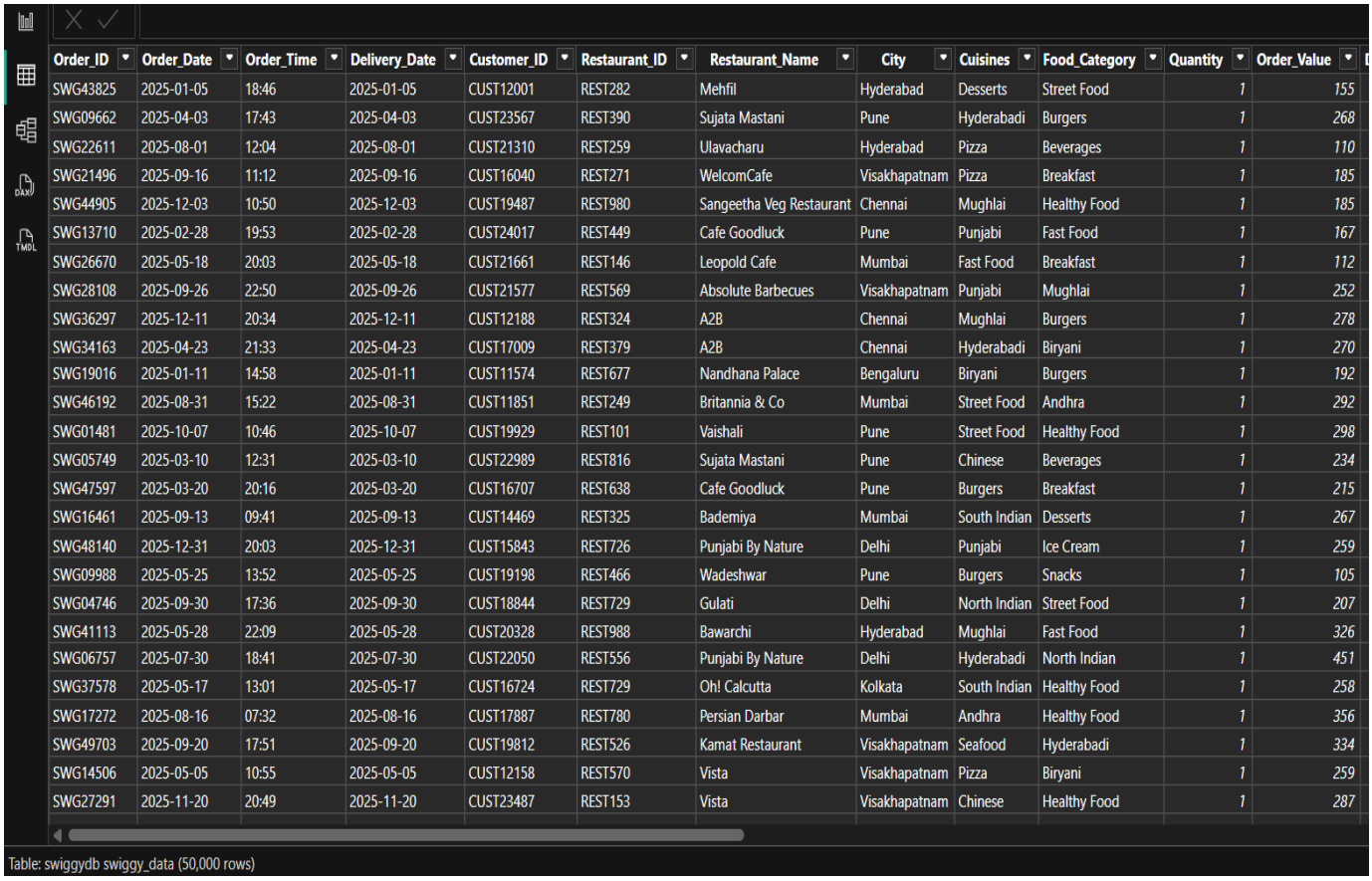
FOOD DELIVERY PERFORMANCE ANALYSIS

A Data Analytics Project Using Microsoft Power BI

Dataset	**swiggy_food_delivery_50000.csv**
Records and fields	50,000 records • 21 columns
Analysis period	01 Jan 2025 – 31 Aug 2026
Tools	Microsoft Power BI, CSV dataset, data profiling



Order_ID	Order_Date	Order_Time	Delivery_Date	Customer_ID	Restaurant_ID	Restaurant_Name	City	Cuisines	Food_Category	Quantity	Order_Value
SWG43825	2025-01-05	18:46	2025-01-05	CUST12001	REST282	Mehfil	Hyderabad	Desserts	Street Food	1	155
SWG09662	2025-04-03	17:43	2025-04-03	CUST23567	REST390	Sujata Mastani	Pune	Hyderabadi	Burgers	1	268
SWG22611	2025-08-01	12:04	2025-08-01	CUST21310	REST259	Ulavacharu	Hyderabad	Pizza	Beverages	1	110
SWG21496	2025-09-16	11:12	2025-09-16	CUST16040	REST271	WelcomCafe	Visakhapatnam	Pizza	Breakfast	1	185
SWG44905	2025-12-03	10:50	2025-12-03	CUST19487	REST980	Sangeetha Veg Restaurant	Chennai	Mughlai	Healthy Food	1	185
SWG13710	2025-02-28	19:53	2025-02-28	CUST24017	REST449	Cafe Goodluck	Pune	Punjabi	Fast Food	1	167
SWG26670	2025-05-18	20:03	2025-05-18	CUST21661	REST146	Leopold Cafe	Mumbai	Fast Food	Breakfast	1	112
SWG28108	2025-09-26	22:50	2025-09-26	CUST21577	REST569	Absolute Barbecues	Visakhapatnam	Punjabi	Mughlai	1	252
SWG36297	2025-12-11	20:34	2025-12-11	CUST12188	REST324	A2B	Chennai	Mughlai	Burgers	1	278
SWG34163	2025-04-23	21:33	2025-04-23	CUST17009	REST379	A2B	Chennai	Hyderabadi	Biryani	1	270
SWG19016	2025-01-11	14:58	2025-01-11	CUST11574	REST677	Nandhana Palace	Bengaluru	Biryani	Burgers	1	192
SWG46192	2025-08-31	15:22	2025-08-31	CUST11851	REST249	Britannia & Co	Mumbai	Street Food	Andhra	1	292
SWG01481	2025-10-07	10:46	2025-10-07	CUST19929	REST101	Vaishali	Pune	Street Food	Healthy Food	1	298
SWG05749	2025-03-10	12:31	2025-03-10	CUST22989	REST816	Sujata Mastani	Pune	Chinese	Beverages	1	234
SWG47597	2025-03-20	20:16	2025-03-20	CUST16707	REST638	Cafe Goodluck	Pune	Burgers	Breakfast	1	215
SWG16461	2025-09-13	09:41	2025-09-13	CUST14469	REST325	Bademiya	Mumbai	South Indian	Desserts	1	267
SWG48140	2025-12-31	20:03	2025-12-31	CUST15843	REST726	Punjabi By Nature	Delhi	Punjabi	Ice Cream	1	259
SWG09988	2025-05-25	13:52	2025-05-25	CUST19198	REST466	Wadeshwar	Pune	Burgers	Snacks	1	105
SWG04746	2025-09-30	17:36	2025-09-30	CUST18844	REST729	Gulati	Delhi	North Indian	Street Food	1	207
SWG41113	2025-05-28	22:09	2025-05-28	CUST20328	REST988	Bawarchi	Hyderabad	Mughlai	Fast Food	1	326
SWG06757	2025-07-30	18:41	2025-07-30	CUST22050	REST556	Punjabi By Nature	Delhi	Hyderabadi	North Indian	1	451
SWG37578	2025-05-17	13:01	2025-05-17	CUST16724	REST729	Oh! Calcutta	Kolkata	South Indian	Healthy Food	1	258
SWG17272	2025-08-16	07:32	2025-08-16	CUST17887	REST780	Persian Darbar	Mumbai	Andhra	Healthy Food	1	356
SWG49703	2025-09-20	17:51	2025-09-20	CUST19812	REST526	Kamat Restaurant	Visakhapatnam	Seafood	Hyderabadi	1	334
SWG14506	2025-05-05	10:55	2025-05-05	CUST12158	REST570	Vista	Visakhapatnam	Pizza	Biryani	1	259
SWG27291	2025-11-20	20:49	2025-11-20	CUST23487	REST153	Vista	Visakhapatnam	Chinese	Healthy Food	1	287

Table: swiggydb swiggy_data (50,000 rows)

1. Executive Summary

This project analyzes a food-delivery dataset containing 50,000 order records and 21 fields. The order dates span 01 January 2025 to 31 August 2026. The analysis focuses on revenue patterns, city and restaurant performance, food-category contribution, delivery operations, and customer ratings. A Power BI report was built around four KPI cards and six visualizations to make these patterns easier to explore.

Across all rows, the dataset contains 50,000 distinct order IDs and ₹26,872,292.18 in recorded Net_Revenue. Average Order_Value is ₹525.58, while the average available customer rating is 4.11 out of 5. The highest recorded revenue by city is Pune (₹3,447,414.25), and the highest-revenue food category is Seafood (₹2,171,568.13).

Interpretation note: the recorded Net_Revenue total includes rows with different order statuses, including cancelled orders. It should therefore be described as recorded revenue in the source data, not automatically as realized or collected revenue. A production analysis should confirm the business rule for revenue recognition and may calculate a delivered-only revenue measure.

2. Business Objectives

- Track recorded net revenue over time and identify high- and low-performing months.
- Compare revenue contribution across cities and food categories.
- Identify the restaurants with the highest recorded revenue.
- Explore the relationship between delivery distance and delivery time.
- Explore how delivery time relates to customer ratings.
- Provide a compact set of KPIs that business stakeholders can filter by year, city, food category, and order status.

3. Dataset Overview

The source file contains 50,000 rows and 21 columns. There are 8 cities, 78 restaurant names, 20 food categories, 15 cuisine labels, and 4 payment methods.

Area	Fields / examples	Purpose
Order and dates	Order_ID, Order_Date, Order_Time, Delivery_Date	Order tracking and time trends
Customer and restaurant	Customer_ID, Restaurant_ID, Restaurant_Name, City	Customer and location analysis
Product / food	Cuisines, Food_Category, Quantity	Demand and category analysis
Financial	Order_Value, Discount, Delivery_Charges, Tax, Net_Revenue	Revenue and order-value analysis
Operations	Order_Status, Delivery_Time, Delivery_Distance	Fulfilment and delivery performance
Feedback / payment	Customer_Rating, Payment_Method	Customer experience and payment mix

4. Data Quality and Preparation

Exact duplicate rows detected: 0.

Customer_Rating contains 3,487 missing values (6.97% of all records). These values should remain missing unless a defensible imputation method is documented; missing ratings should not be treated as zero.

The Order_Status distribution is: Delivered 43,953 (87.91%), Cancelled 3,487 (6.97%), and Delayed 2,560 (5.12%).

There are 3,487 rows with Delivery_Time equal to zero, matching the count of cancelled rows in this file. This pattern suggests zero may be used as a placeholder for cancelled orders rather than a real delivery duration. For delivery-time analysis, exclude cancelled orders and validate how delayed orders are defined.

Recommended preparation steps in Power BI: convert Order_Date and Delivery_Date to Date types; derive Year, Month_Number, Year_Month, and a display month label; sort the display month by Year_Month; check numeric field types; and verify that the revenue measure matches the intended business definition. Keep the rating blanks as blanks.

5. KPI Definitions and Results

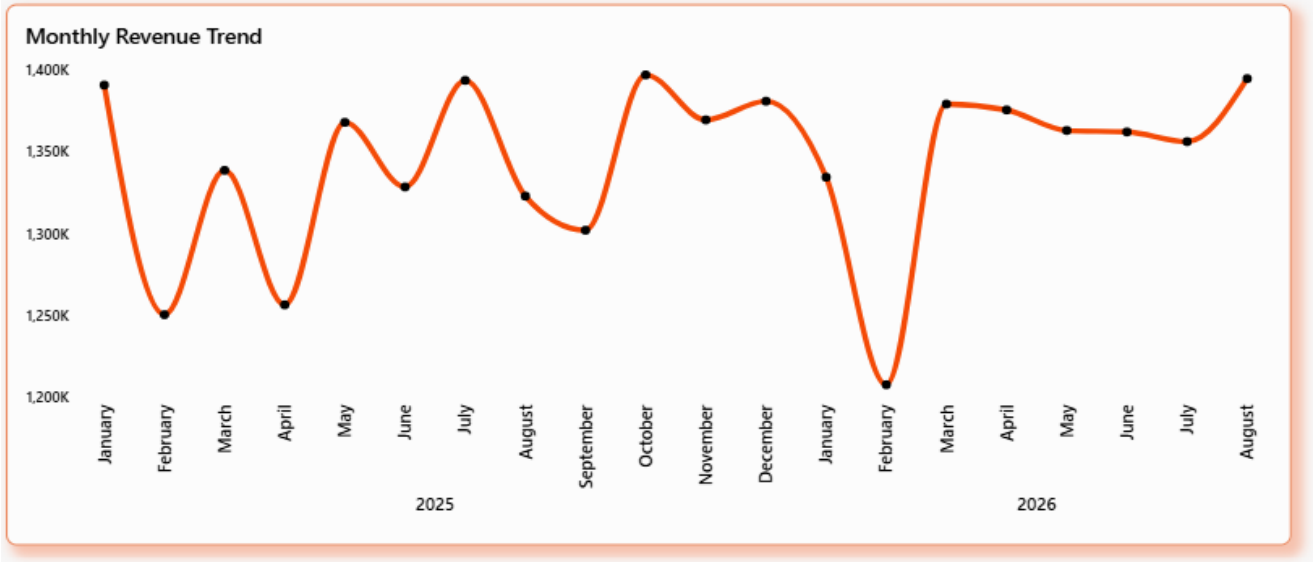
KPI	Definition	Dataset result
Total Orders	Distinct count of Order_ID	50,000
Total Recorded Net Revenue	Sum of Net_Revenue across all rows/statuses	₹26,872,292.18
Average Order Value	Average of Order_Value	₹525.58
Average Customer Rating	Average of non-missing Customer_Rating	4.11 / 5

For context, delivered orders alone total 43,953 and their recorded Net_Revenue sums to ₹23,591,236.98. The average delivery time among delivered orders is 34.89 minutes, and the average rating among delivered orders with ratings is 4.16 / 5. These delivered-only figures are useful alternatives if the dashboard is intended to represent completed service.



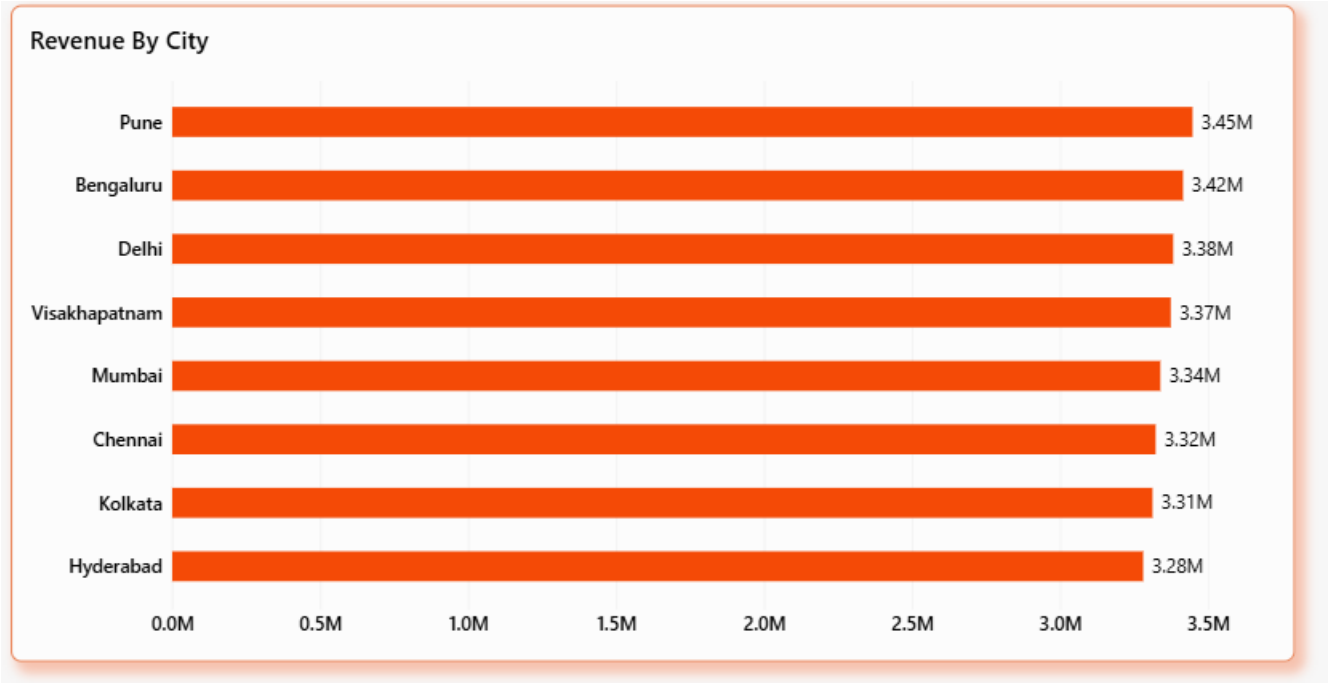
6. Dashboard Visualizations and Findings

6.1 Monthly Revenue Trend



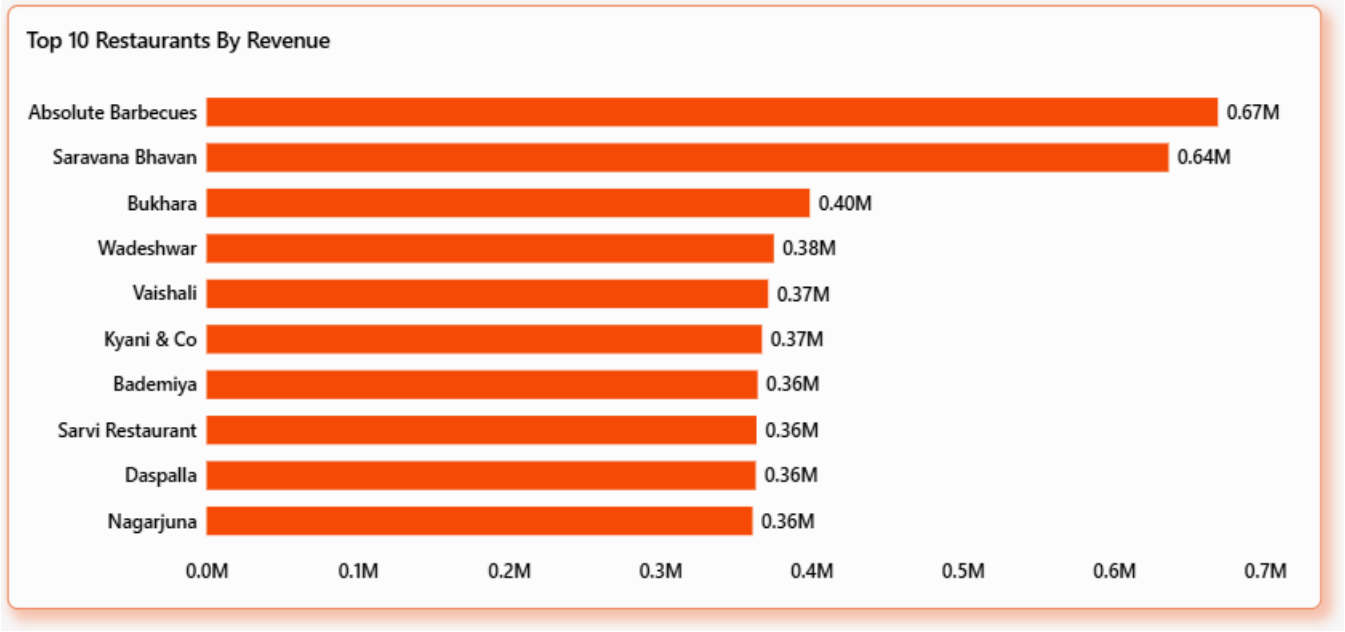
The month with the highest recorded Net_Revenue in the available period is 2025-10 (₹1,396,906.86). Monthly values should be interpreted in context: the dataset contains a partial 2026 year, so year-over-year comparisons should use equivalent months and the same revenue definition.

6.2 Revenue by City



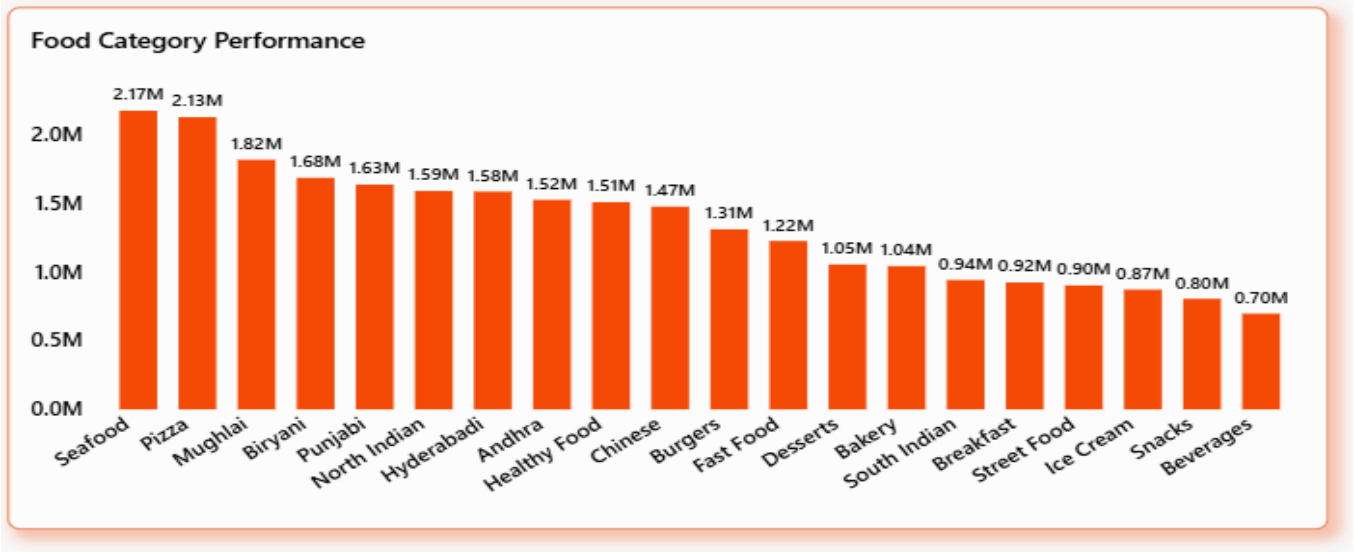
Pune has the highest recorded Net_Revenue at ₹3,447,414.25. Hyderabad has the lowest at ₹3,280,452.21. The difference between the two is ₹166,962.04. The city ranking is relatively close, so consider also comparing order counts, average order value, and delivered-only revenue.

6.3 Top 10 Restaurants by Revenue



Absolute Barbecues ranks first by recorded Net_Revenue (₹668,689.34), followed by Saravana Bhavan (₹636,127.73). Restaurant performance should be checked against order volume and order status to distinguish high order volume from high average order value.

6.4 Revenue by Food Category



Seafood is the leading category by recorded Net_Revenue (₹2,171,568.13), followed by Pizza (₹2,125,396.74). Category revenue can help guide menu placement and promotional tests, but decisions should also account for margin, discounts, and order counts.

6.5 Delivery Distance vs Delivery Time



The Pearson correlation between Delivery_Distance and Delivery_Time across available rows is 0.57, indicating a positive association in this dataset. This is consistent with longer trips often taking longer, but it does not prove causation. Traffic, weather, restaurant preparation time, and rider availability are not represented in this simple comparison.

6.6 Delivery Time vs Customer Rating



The correlation between Delivery_Time and Customer_Rating for records with both values is -0.10, a weak negative association. This suggests ratings tend to be slightly lower as delivery time increases, but delivery time alone does not explain ratings. Ratings are missing in {missing_ratings:,} records, so this relationship reflects only rows with a recorded rating.

7. Key Findings

- Recorded revenue is ₹26,872,292.18 across 50,000 distinct orders, using all order statuses.
- Pune leads the city comparison with ₹3,447,414.25 in recorded Net_Revenue.
- Seafood is the top food category by recorded Net_Revenue.
- Absolute Barbecues is the highest-revenue restaurant in the dataset.
- Delivery distance and delivery time have a positive correlation (0.57).
- Delivery time and customer rating have a weak negative correlation (-0.10).
- Customer ratings are missing for 3,487 records, and cancelled rows appear to use zero delivery time.

8. Business Recommendations

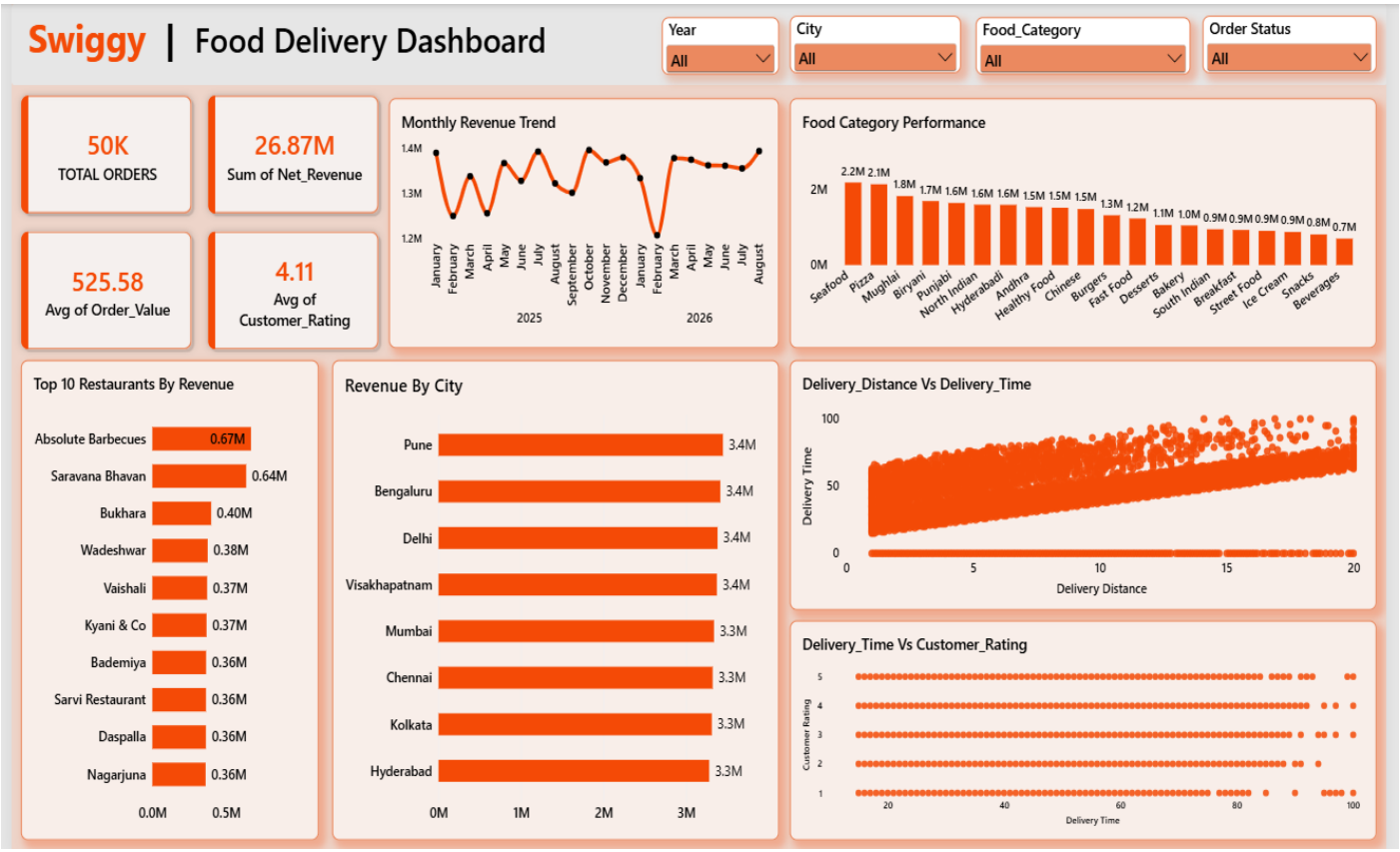
- Confirm the revenue-recognition rule. Use a delivered-only revenue measure for completed-order performance, and show cancelled / delayed order metrics separately.
- Investigate why 3,487 orders are cancelled and whether cancellation rates vary by city, restaurant, payment method, or time of day.
- Review delivery-time performance by distance bands and city; compare the median as well as the average to reduce the effect of extreme values.
- Use customer ratings with care: retain missing values as missing, and analyze ratings alongside order status and delivery time.
- Run targeted menu or promotion experiments for leading categories such as Seafood, while monitoring margin and discount cost.
- Add operational KPIs such as delivery success rate, cancellation rate, delayed-order rate, and delivered-only average delivery time to future dashboard iterations.

9. Limitations

- The dataset may be synthetic or sample data; findings should not be represented as verified real-world Swiggy business results unless the data provenance is confirmed.
- Revenue values exist on cancelled rows, so total Net_Revenue across all statuses may not equal revenue actually collected.
- Customer_Rating has missing values, and Delivery_Time equals zero for a set of rows matching the cancelled-order count.
- The dataset does not include key explanatory factors such as traffic, weather, restaurant preparation time, delivery partner availability, or profit margin.
- The analysis describes associations and rankings; it does not establish causal effects.

10. Conclusion

This project demonstrates an end-to-end business reporting workflow: profiling a structured dataset, defining KPIs, building Power BI visualizations, comparing performance by month, city, restaurant, and food category, and exploring operational relationships. The most important next improvement is to validate the revenue definition and ensure cancelled orders and missing operational values are handled consistently. With these checks documented, the report can serve as a useful portfolio example of analytical thinking, dashboard design, and data-quality awareness.



Appendix: Dashboard Checklist

- Line chart — Monthly Revenue Trend (Year_Month_Display sorted by Year_Month).
- Bar chart — Revenue by City.
- Bar chart — Top 10 Restaurants by Revenue.
- Column chart — Revenue by Food Category.
- Scatter chart — Delivery Distance vs Delivery Time (Order_ID in Details).
- Scatter chart — Delivery Time vs Customer Rating (Order_ID in Details).
- KPI cards — Total Orders, Total Recorded Net Revenue, Average Order Value, Average Customer Rating.
- Slicers — Year, City, Food Category, Order Status.